

History of the protocol stack

The early days of the Web: The 1990s

As indicated earlier, the early version of the Web consisted of the stack shown in Figure 3.

The initial implementation of the Web protocol stack was founded on four key protocols: DNS, IP, TCP and HTTP. While the other three protocols have remained largely like what they were then, HTTP has followed a slightly different path. The early Web used HTTP 1.0, which was defined in May 1996 but was revised as HTTP 1.1 in 1999. HTTP 1.0 had the disadvantage of permitting only a single request-response pair on a TCP connection, which was addressed in HTTP 1.1. This and other improvements made HTTP 1.1 the Internet workhorse for a long period of time.

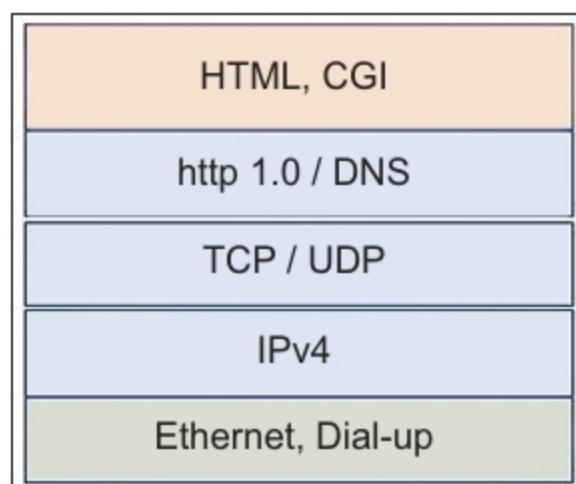


Figure 3: Protocol stack in the 1990s

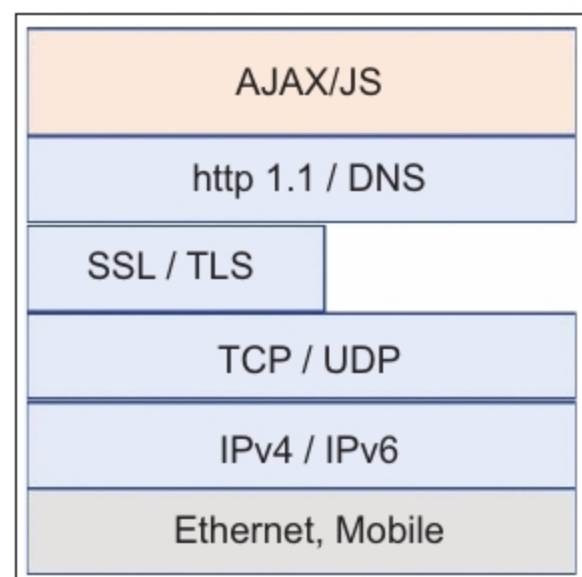


Figure 4: Protocol stack in the first decade of the 2000s

Web 2.0: The first decade of the 2000s

As the Internet grew, it became a platform suitable for commercial activities, primarily because it brought together many parties. Applying the Web for commerce naturally led to concerns of message integrity, confidentiality and authenticity. HTTP, being a simple text-based protocol, offered none of these. Thus was born Secure Socket Layer or SSL. This protocol also evolved quickly, and by the early 2000s, SSLv3 was good enough security for that time. This added a new layer to the Web stack, as shown in Figure 4. However, it was an ‘optional’ layer and protocols like DNS continued to be in clear text (just to reiterate, it was not so much about deployment but rather about definition — DNS over SSL was not defined till very recently).

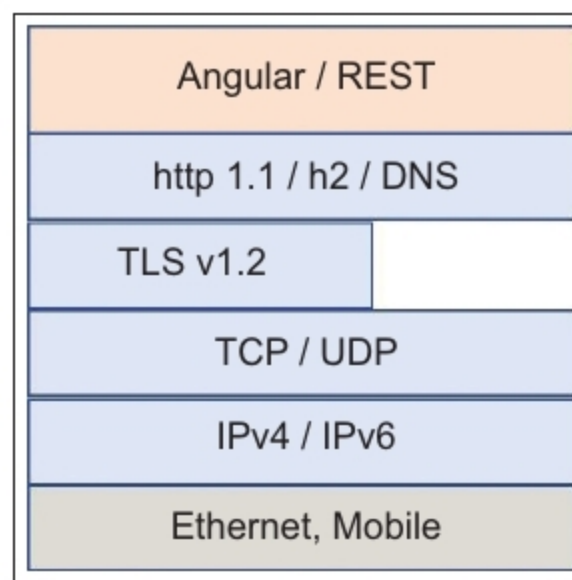


Figure 5: Protocol stack circa 2010-2015

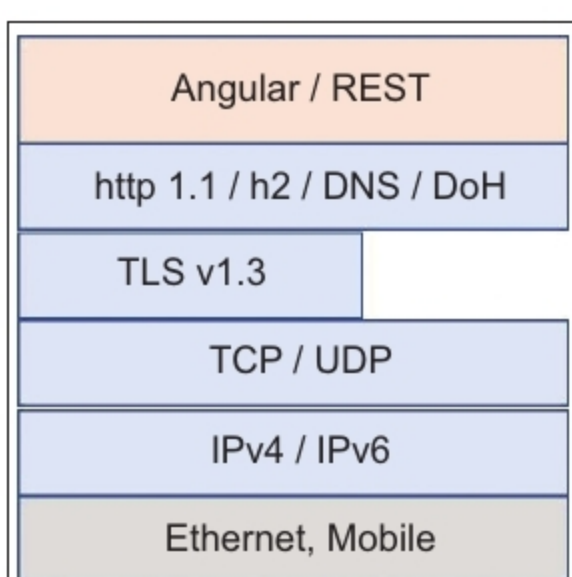


Figure 6: The stack with emerging protocols

If the early Web took off in a big way, Web 2.0 was much more amenable for developing a variety of Internet based applications, including maps, online shopping, social networking, and so on. This stack quickly became the pillar for most Internet communications.

Further advances

The initial protocol stack shown in Figure 2 was the framework for the Web 2.0 advances that happened in the first decade of the new millennium. Separately, the growth of the Internet drove other activities, which are summarised below.

1. The security of Web transactions continued to be a concern, and the community responded by developing three versions of TLS: 1.0, 1.1 and 1.2.
2. Concerns about the paucity of IPv4 addresses led to the initial deployments of IPv6, but the latter did not take off in a big way as organisations found ways to manage with the scarce IPv4 addresses, especially by using techniques like NAT.
3. 3G, and then LTE networks, were built and deployed, which started an entire new stream of Web clients.
4. Near the end of the first decade of 2000, social networking tools emerged, leading to users demanding an ‘always-on’ experience.

All these changes led to two major concerns or requirements:

- Reduced latency for every kind of device from anywhere in the world.
- Improved security and privacy for everyone.

Google, primarily because of its global reach and business, took the lead in attacking the first of the two problems. Acting from extensive data collected by the Chrome/Google server end-to-end systems, Google researchers proposed SPDY as an alternative to HTTP 1.1. The overall proposal was for optimisation, either by cutting bits in the headers (by making them binary and compressing them) or by cutting wait time by allowing concurrent request-responses on a single connection. Eventually, SPDY was taken to IETF